DATA CLEANING AND VISUALIZATION PROJECT

Thiranex EduTech LLP - Internship Task 1

1. Introduction

This project demonstrates data cleaning and exploratory data analysis using Python. A synthetic employee dataset was created for this internship task.

2. Objective

To identify and handle missing values, duplicate records and outliers, and understand employee data through statistical analysis and visualizations.

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook and ReportLab.

4. Dataset Description

The synthetic dataset initially contained 20 employee records and six columns: Name, Age, Salary, Department, Gender and Join_Date.

5. Data Cleaning

Missing Age values were filled using the median. The missing Salary value was filled using the median. Text values were standardized, duplicate names were removed, and salary outliers were handled using the IQR method. Join_Date was converted into datetime format. Department and Gender were encoded using One-Hot Encoding.

6. Statistical Findings

Total employees after cleaning: 18
Average age: 29 years
Average salary: Rs. 57,923.98
Highest salary: Rs. 1,10,000
Lowest salary: Rs. 30,000
Department employee counts: IT - 6, HR - 5, Finance - 4, Sales - 3.

7. Key Insights

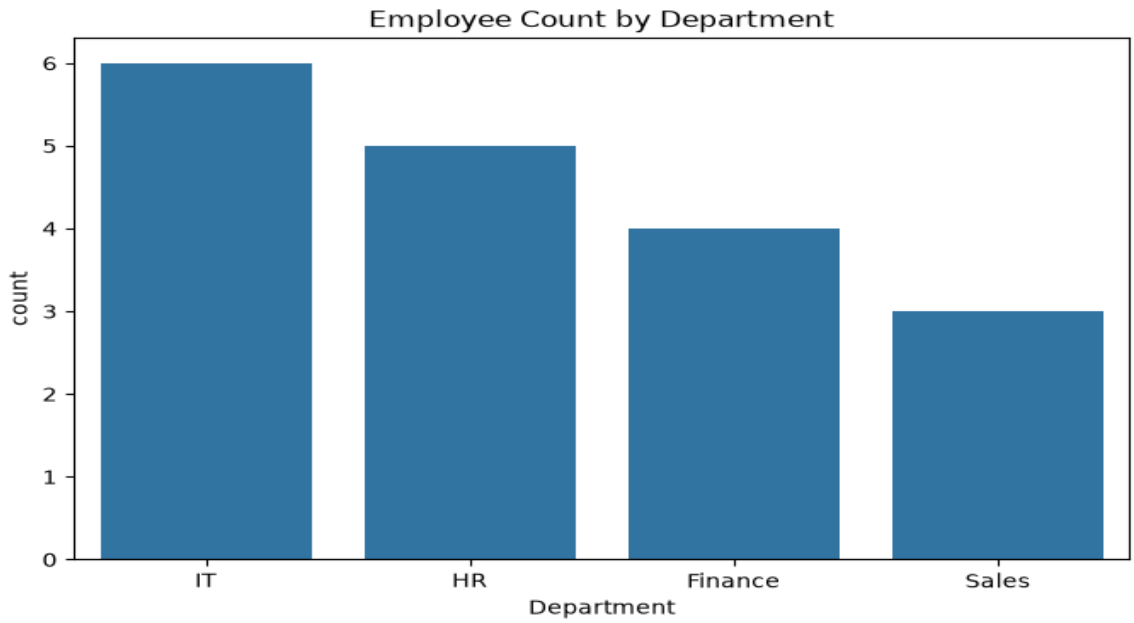
IT had the highest employee count and highest average salary (Rs. 61,166.67). Sales had the lowest average salary (Rs. 50,000). The analysis also explored salary distribution and the relationship between age and salary.

8. Conclusion

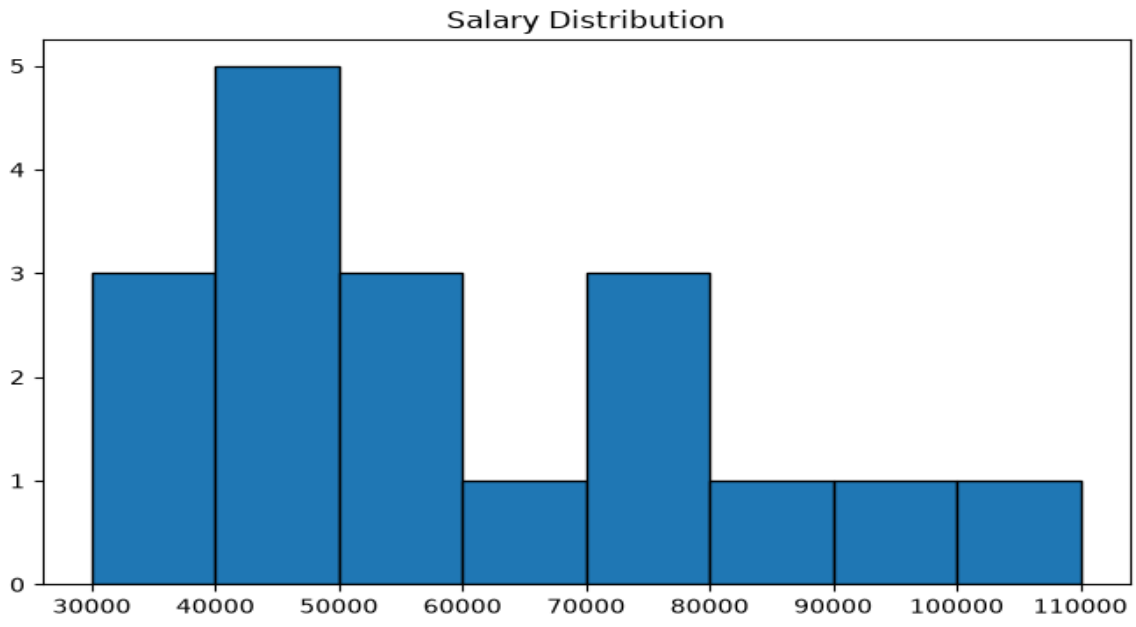
The project successfully demonstrated data cleaning, preprocessing and exploratory data analysis using Python. Visualizations and statistical summaries helped identify patterns in the synthetic employee dataset.

9. Data Visualizations

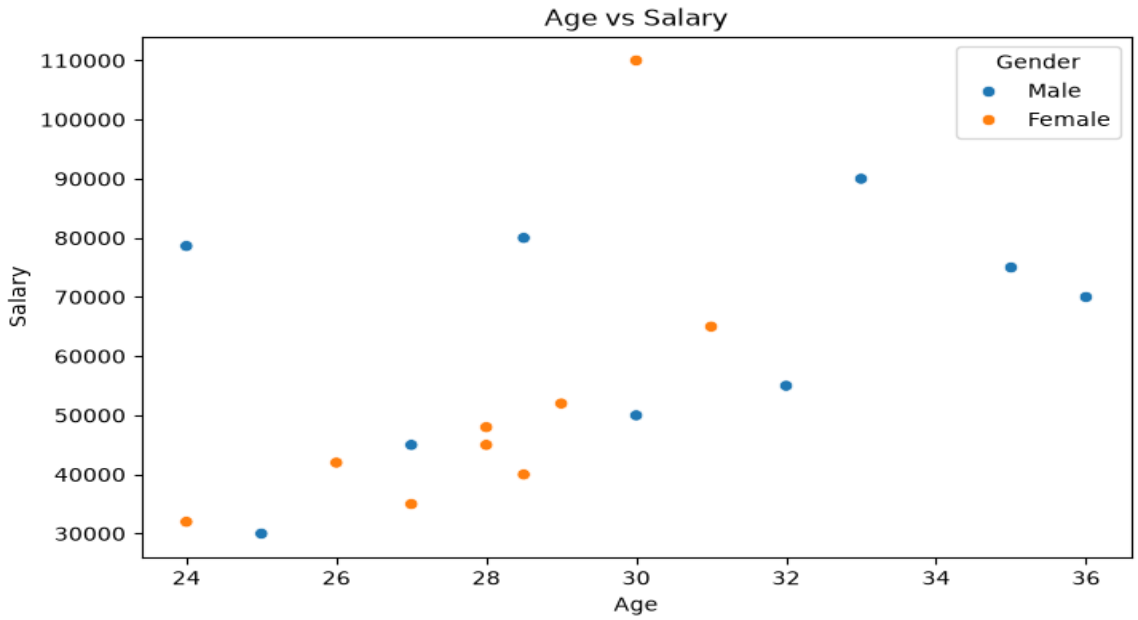
Employee Count by Department



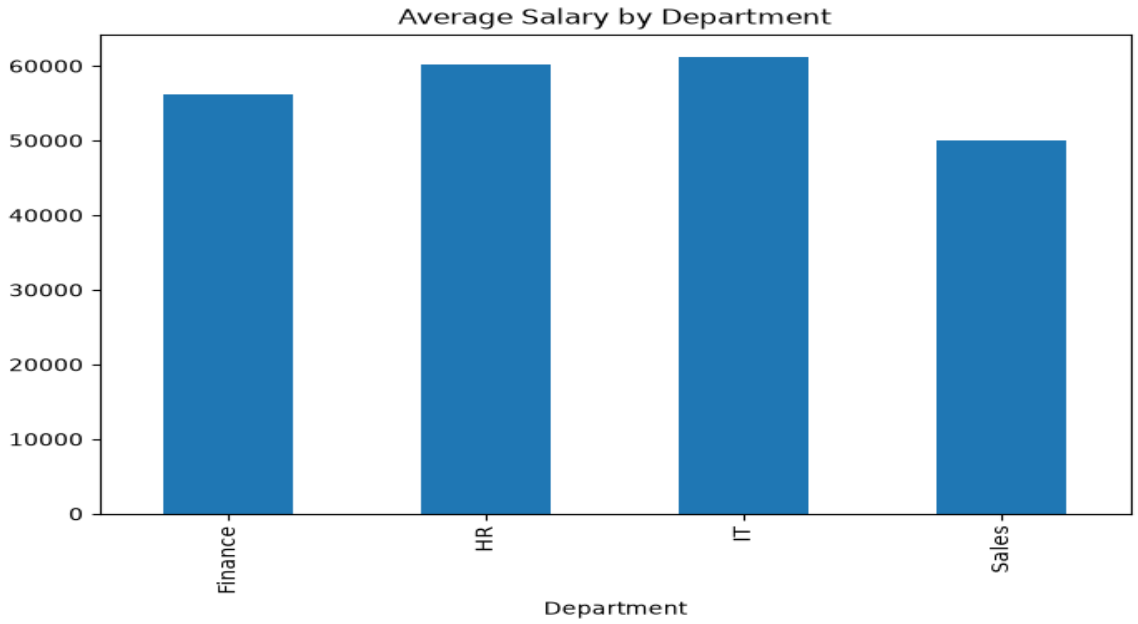
Salary Distribution



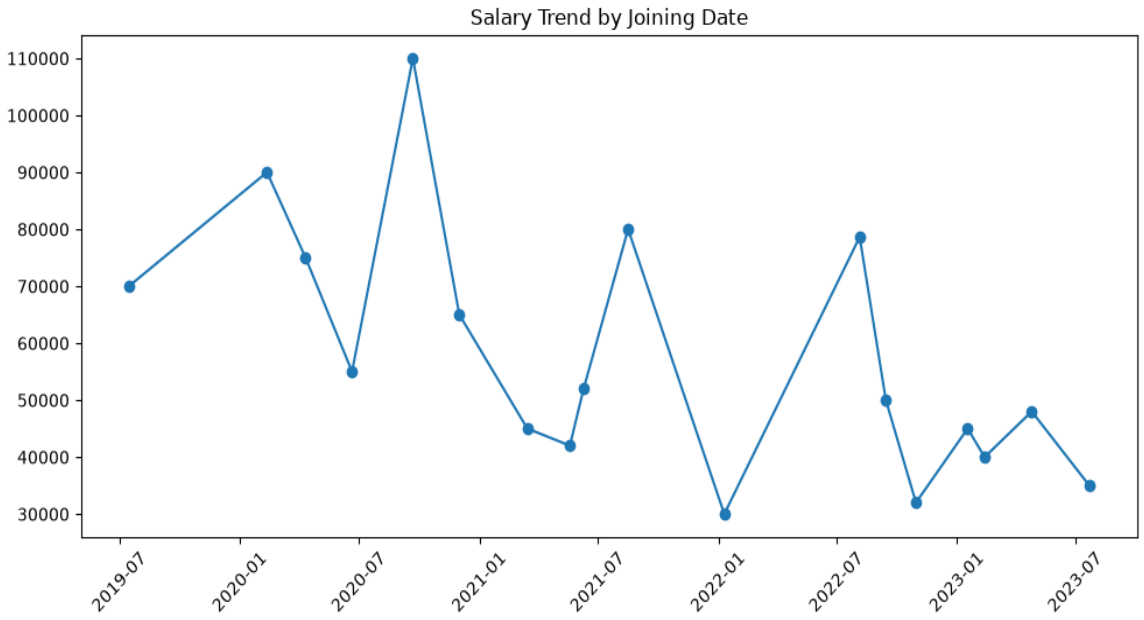
Age vs Salary



Average Salary by Department



Salary Trend by Joining Date



Correlation Heatmap

